COMPETITION PITCH DECK

PulseRank *AI*

THE REASONING RECRUITER FOR THE POST-RESUME ERA

Traditional hiring systems treat candidates as static bags of words. PulseRank AI maps multi-axis engineering signals and runs adversarial reasoning tournaments to rank talent accurately.

Why Traditional Systems Are Broken

- **Keyword Stuffing Bias:** Traditional ATS filters reward candidates who copy-paste exact buzzwords, creating a high volume of false positives.
- **Rigid Boolean Matching:** Literal string checks miss perfect adjacent talent (e.g., a "distributed systems" developer is highly qualified to build "fraud pipelines").
- **Neglected Career Trajectory:** Typical resumes are parsed for titles and durations in isolation, ignoring promotions, velocity, and project impact.
- **No Risk Screenings:** Candidate engagement decline, sudden job hopping, and seniority mismatches go unnoticed until recruiter phone screen stages.

The Recruiter Reasoning Tournament

1. Graph Parser

Parses raw job descriptions into structured node graphs mapping core must-haves, preferred adjacent skillsets, and latent domain concepts.

2. 8-Factor Scorecard

Evaluates candidate profiles across 8 distinct dimensions including semantic fit, career velocity, project impact, learning agility, and risk profiles.

3. Adversarial Bracket

Executes Chess-style head-to-head pairwise matches across candidates, outputting stable ELO ratings that adjust dynamically to JD weight changes.

Pairwise ELO & Matchup Engine

Adversarial Decision Model

When two candidates are compared, if the scorecard score delta is greater than `4` points, the higher score wins. If ≤ `4` , contextual tie-breakers resolve the match:

- **Senior Roles:** Prioritizes system ownership evidence and mentorship.
- **Mid/Junior Roles:** Prioritizes learning agility and career velocity.

Performance & Scale

The engine runs in `O(N log N)` time complexity. It sorts 1,000+ candidate profiles in under 13 seconds.

Chess-style ELO ratings are updated iteratively over 3 simulation tournament rounds using a standard K-factor of `32` .

Multi-Axis Scorecard Pillars

1. Semantic Fit
Measures context-aware matching of skills using semantic concept mapping.

2. Seniority Alignment
Calibrates experience scope, title tier, and expected team responsibilities.

3. Career Velocity
Measures promotion frequency and timeline growth rate.

4. Project Impact
Extracts and scores quantifiable metrics (e.g., Millions of users, throughput scales).

5. Learning Agility
Scores recent tech acquisitions and open-source contributions.

6. Initiative
Looks for self-driven builds, major refactors, and ownership.

7. Culture Alignment
Identifies leadership, mentorship, and active team collaboration indicators.

8. Risk Analysis
Screens for warning signs like frequent job-hopping and low response rates.

Recruiter Workspace Console

- **Dynamic Weighting Sliders:** Tune signal weights (semantic fit, velocity, risk) in real time to immediately recalculate ELO scores and brackets.
- **Interactive SVG Graph:** Real-time visualization of parsed Job Description skills and adjacent concepts.
- **Deep Scorecard Inspector:** Click any candidate to view their strengths, risk flags, transferable skills, and detailed background timelines.
- **Shortlist Export:** One-click shortlist creation with direct download as a clean, standardized CSV.

High-Performance Tech Stack

Frontend Core

Built with **React** & **TypeScript** using **TanStack Router** for file-based routing and lightning-fast developer experience.

Styling System

Custom Vanilla CSS built with tailored HSL/OKLCH color variables to render the premium cream paper and vermillion interface.

Data Viz

Uses **Recharts** for analytical breakdowns and dynamic SVG rendering for the real-time parsed requirement networks.